

Development of the 'Spot' Mobile Application to Improve Cleanliness of Commercial Spaces in Mérida, Yucatán

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Abstract

This paper documents the user-centered design process of **Spot**, a crowdsourcing and collaborative economy mobile platform aimed at addressing solid waste accumulation and toxic leachate (*pasta negra*) in the commercial zones of Mérida, Yucatán. Leveraging HCI and UCD principles, the project bridges the service gap left by public utilities. The study covers user profiling (personas and scenarios), requirements engineering (SRS v1.0 — 17 functional and 6 non-functional requirements), and the development of low- and high-fidelity prototypes validated through a structured field usability testing plan with five participants.

Description

Spot resolves the “responsibility gap” — where neither city hall nor passersby assume immediate waste cleanup — through a three-phase collaborative workflow connecting citizens, merchants, and autonomous cleaning brigades:

- 1 Report & Alert** — A citizen or merchant photographs waste accumulation, submits a geotagged report, and the system auto-extracts GPS coordinates and alerts nearby merchants in real time.
- 2 Offer & Match** — The merchant publishes a cleanup request with a micro-payment offer (00–00 MXN). Available brigades see geo-ranked jobs on an interactive map and accept the task.
- 3 Execute & Validate** — The brigade attends the site (verified by geofencing), performs cleanup using biodegradable chemicals, and submits before/after photo evidence. The merchant approves and releases digital payment.

Two core technologies underpin the platform: **geofencing**, which verifies physical attendance of the brigade without direct human supervision, and **gamification** (leaderboards, ecological badges, and point systems), which incentivizes sustained citizen participation and brigade loyalty.

Proposed architecture for the Spot collaborative platform

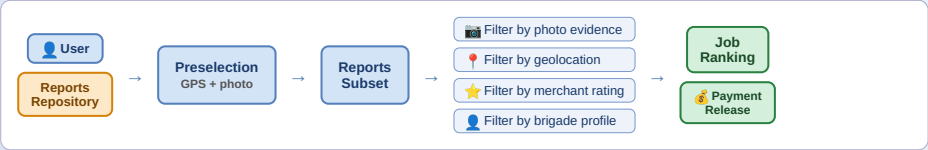


Fig. 1. Proposed architecture for the Spot hybrid collaborative platform

Spot graphical user interface

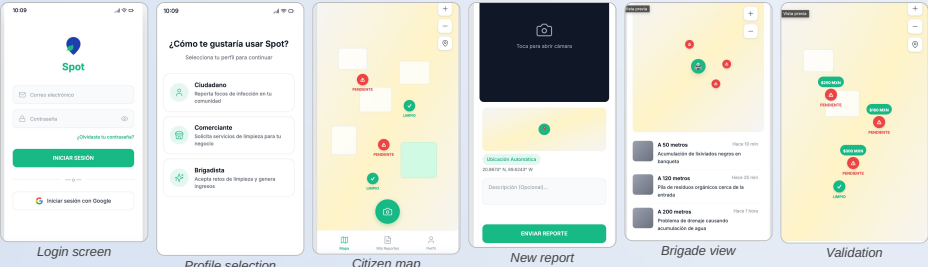


Fig. 2. Spot graphical user interface — complete application flow

An example of the impact dimensions addressed by Spot



Fig. 3. Three dimensions of socio-ecological impact of the Spot platform in Mérida, Yucatán



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